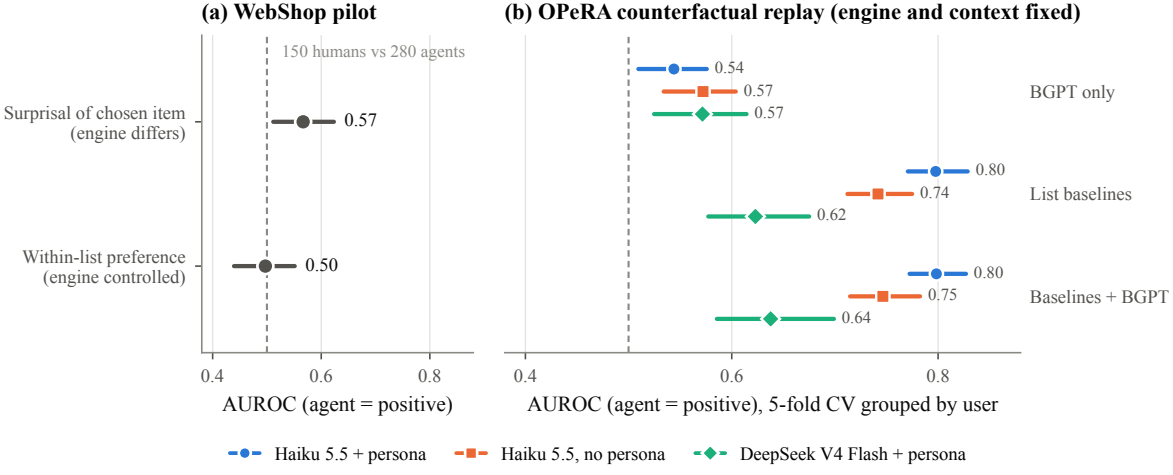
(a) WebShop pilot
150 humans vs 280 agents
Surprisal of chosen item
(engine differs)
0.57
Within-list preference
(engine controlled)
0.50
0.4
0.6
0.8
AUROC (agent = positive)
(b) OPeRA counterfactual replay (engine and context fixed)
0.54
0.57
0.57
BGPT only
0.80
0.74
0.62
List baselines
0.80
0.75
0.64
Baselines + BGPT
0.4
0.6
0.8
AUROC (agent = positive), 5-fold CV grouped by user
Haiku 5.5 + persona
Haiku 5.5, no persona
DeepSeek V4 Flash + persona